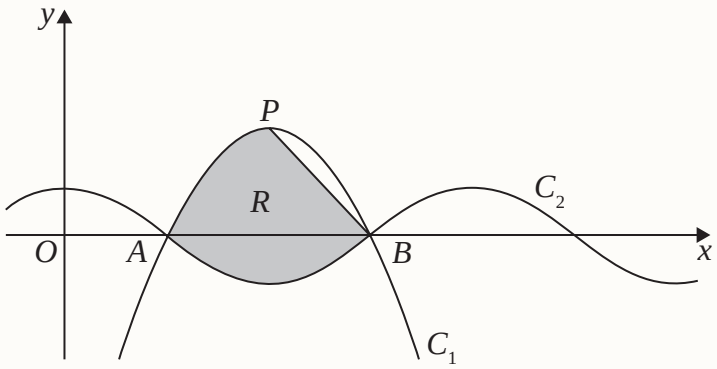



**Figure 2**

(a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where $a$, $b$ and $c$ are constants to be found.

**(3)**

Figure 2 shows part of a sketch of curve $C_1$ with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point $P$ is the maximum point on $C_1$

(b) state the coordinates of $P$

**(2)**

Figure 2 also shows part of a sketch of curve $C_2$ with equation

$$y = \cos(kx)$$

where $k$ is a constant and $x$ is measured in radians.

Given that $C_1$ and $C_2$ intersect on the $x$-axis at point $A$ and at point $B$, as shown in Figure 2,

(c) (i) state the $x$ coordinate of $B$

(ii) state the value of $k$

(iii) state the period of $C_2$

**(3)**

The line segment $L$ joins $P$ and $B$.
The region $R$, shown shaded in Figure 2, is bounded by $L$, $C_1$ and $C_2$

(d) Use inequalities to define $R$.

**(5)**